# Additional Specific Recommendations for Scoring Well on Lab Practical

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1. When you get to the station, read the text to your Version (A, B, or C). The versions are color-coded: green, yellow, purple. KNOW YOUR COLOR. It is the color of the paper you will use.
2. Knowing the Time Limit:
   You have 4 minutes to read the instructions and carry out the task. The task may be to use a device of the lab in some way (not always).
   - A station (or two) may ask you to determine a mass on the electronic balance. You should know how to tare (zero) a balance by now, and to record the mass
   - A station may ask you to do a volumetric determination on a volumetric container, like a buret, eudiometer, graduated cylinder, etc. You should know about the meniscus AND you should know the PRECISION of the device/instrument
   - A station may ask you to read another measured quantity, such as pressure or a length with a ruler!

   Because of the time limit, make sure you at least record the data/info of your station in the box so that you can use that information later to do a calculation within the 4 min time limit. You are able to finish calculations for one station if you have extra time remaining after doing another station, and at the end of the practical rotations.

3. QUANTITIES = numbers with units
   Where students lose points is failing to use the proper units for the number you will give as a result. If you are asked to give a *molar mass*, then the student should know those always

| ***Dimension*** | ***Usual or Typical Units*** |
|---|---|
| **Mass** | g(rams) |
| **Volume** | L(iters) for non-solids, $cm^3$ for solids usually, 1 mL = 1 $cm^3$ |
| **Molar Mass** | g/mol (grams per mole) [compare to mass on atomic scale] |
| **Pressure** | atm, mmHg, torr, (less so: psi, $N/m^2$, Pascal) |
| **Density** | g/mL, $g/cm^3$ |
| **Temperature** | K (Kelvin), °C (Celsius); K=273 + C, C = K-273 |
| **Mass on atomic scale** | amu (atomic mass units) when the mass is about a single atom or molecule (compare to molar mass) |
| **Powers of 10 name prefixes** | • kilo=1000, $10^3$<br>1 kilogram (kg) = 1000 g = $10^3$ g<br>1 kiloliter (kL) = 1000 L = $10^3$ L (rare usage)<br>• milli=1/1000th, $10^{-3}$<br>1 milligram (mg) = 0.001 g = $10^{-3}$ g<br>1 milliliter (mL) = 0.001 L = $10^{-3}$ L |

4. PRECISION: significant digits and decimal places
   Throughout the semester you have been scored on your use of the proper significant digits and of decimal places. Remember the rules. The rubric for the lab practical includes getting points for the correct use of significant digits and decimal places.
5. CALCULATIONS
   ***SHOW YOUR WORK! SHOW YOUR WORK! SHOW YOUR WORK!***
   It cannot be sufficiently emphasized that students will lose points in making a calculation and writing in the result only, and NOT showing the mathematical expressions. Failure to show work

can even be points deductions. Write out what you are thinking. It will be worth partial credit and demonstrate your mind using chemistry.

6. You will only be asked about features of the experiments you did, not other experiments. Follow the study guide for those experiments that you did.

Your instructor does not produce the questions and setups of the Lab Practical, but the instructor does score it against a rubric and determines your final score.

The practical is worth 100 points, but that 100 points is worth FIVE PERCENT (5%) of the course grade. Canvas Gradebook does the calculation of the weight, so it has MORE value than 100 points of a Homework/ or any MOM assessment! That is 100 points on the Practical is NOT EQUAL to 100 points of MOM assessments (homework, quizzes!!!)

The same is true for the Course Final exam. It may be scored from 0 to 100 percent (100%), but it is worth 12.5% of your grade potentially